



SADP20 Pipette Manual

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1. Product Overview

SADP (Air Displacement Pipette) is a pipetting module developed by Keyto based on aspirate and dispense technology with tips to avoid contamination of samples and pipelines. The module includes a highly sensitive pneumatic liquid level detection (pLLD) that can detect the liquid level very quickly.

1.1 SADP20 Series Pipettes Main Features

① This pipette has an integrated drive controller that controls the motor operation by receiving commands from the host computer. This enables fast aspiration and dispense of liquid. Serial protocols include RS232, RS485 and CAN bus.

② This pipette is driven by a through-shaft linear motor for precise aspirating and discharging at high velocity.

③ Advanced air cylinder processing technology, strict inspection procedures, and unique sealing solutions ensure that the air cylinder's service life can reach more than 1 million cycles without maintenance.

④ Good chemical resistance and high wear resistance of the materials of the parts in contact with the disposable tips. It ensures that the disposable tips cone do not need to be replaced during their life cycle.

⑤ A variety of syringes with capacities from 100 μ L to 1mL can be allocated.

⑥ End position optical detection. The optical sensor detects whether the pump is at the initial point of aspirate.

1.2 Definition

Accuracy f: It refers to the deviation degree between the result of measurement and actual value. It can be calculated as the following formula.

$$f_s = \frac{X_s - X}{X_s} \times 100\%$$

f_s — accuracy.

X — the measurement result.

X_s — the actual value.

Coefficient of variation (CV): Coefficient of variation also known as relative standard deviation, is a standardized measure of dispersion of a probability distribution or frequency distribution.

$$CV = (\text{Standard Deviation SD} / \text{Average value MN}) \times 100\%$$

2. Product Features

2.1 How to Order

①	②	③	④	⑤	⑥	⑦
	P		-	-	/	

Note code	Note type	Name	Product naming instructions
①	Pipette type	S	Single channel
②			Air Displacement Pipettor
③	Motor size	20	20mm
④	Function & features	2	with controller
⑤	Model	0	No model difference
⑥	Displacement	100/250/010	100ul/250ul/1000ul
⑦	Other requirements	—	—

Notes:

1. The naming rules of this product contains only the pipette, not include the Z-axis.
2. Please contact our company for customizing Z-axis.

SADP20 Single aspirate single dispense

Disposable tips volume/ul	Dispense volume/ul	Dispense CV
50(with filter)	5	3.0%
50	10	2.0%
200(with filter)	10	3.0%
200	50	1.0%
1000	10	5.0%

1. The test environment is 21-25℃ static wind environment;

2. The test mode is non-contact liquid discharge method;

3. single suction and single discharge is to replace the new TIP head after each test;

4. When the liquid discharge does not reach the ideal discharge volume, it is necessary to increase the compensation value;

SADP20 Single aspirate multi-dispense

Disposable tips volume/ul	Dispense volume/ul	Dispense CV
1000(with filter)	20	3.0%
1000	50	2.0%
200	10	5.0%

1. The test environment is 21-25℃ static wind environment;

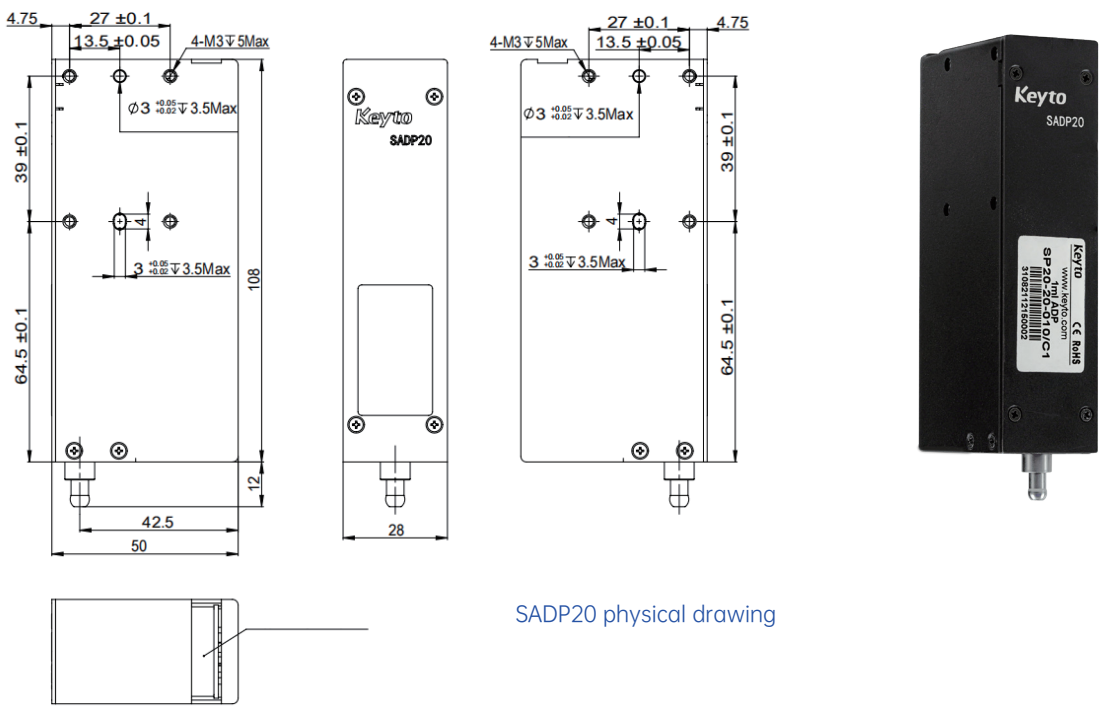
2. The test mode is non-contact liquid discharge method;

3. single suction and single discharge for replacing the new TIP head after each test;

2.2 SADP20 Series Pipettes Specifications

Specification	SADP20-2 Pipette		
Maximum dispense volume(μL)	100	250	1000
Liquid volume per step(ul/step)	0.06	0.15	0.6
Lifetime	100		
Working pressure (MPa)	< 0.06		
Drive communication protocol	RS232, RS485, CAN bus optional		
Suitable disposable tip types	Compatible with dominate brand, can be customized according to customer specific needs		
Medium	Air		
Weight (g)	< 400g		

2.3 SADP20 Series pipette shape and mounting dimensions



SADP20 physical drawing

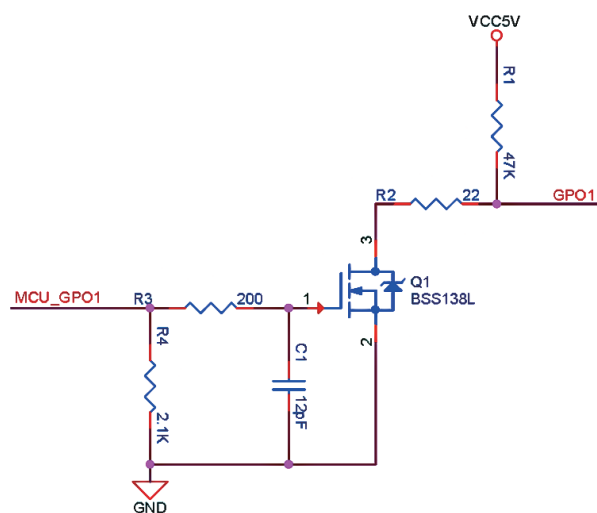
3. Product Electric Interface



The SADP20 is connected to the external control system using highly flexible towline cable AWG26, the following chart are definition of communication cable.

Function	Cable color	Remarks
DC 24V+	Red	Power input $\pm 1\%$, $\geq 1A$
GND	Black	Ground
RS232-RX	Green	Communication interface
RS232-TX	Blue	Communication interface
RS485A	Orange	Communication interface
RS485B	White	Communication interface
CANL	Yellow	Communication interface
CANH	Yellow-green	Communication interface
GPO1	Purple	pLLD digital output, 5V voltages
GPO	Brown	Spare

The GPO1 interface circuit is as follows.



4. Communication Protocols

4.1 Communication Method

• Communication interface

The following methods are supported for communication with the SADP20. Default baud rate:

- ◆ RS232
- ◆ RS485
- ◆ CAN
- ◆ Serial interface: 38400
- ◆ CAN: 500K

• Protocols

SADP supports serial interface and CAN communication, where the serial interface is divided into RS232 and RS485, both of which have the same communication protocols.

- ◆ DT Protocol
- ◆ OEM Protocol
- ◆ CAN Protocol

• DT Protocol

The protocol is based on RS232 and RS485 communication, the communication protocol has no check bit, you can easily use the serial interface debugging tool to send a string to control the SADP. Data sent and received as ASCII string, which is convenient for users to debug. The communication protocol has no check bit so there is a risk of data loss, please see DT protocol for details.

• OEM Protocol

The protocol is based on RS232 and RS485 communication, the communication protocols include communication sequences and check bit, which can effectively prevent data loss. During the working period, the controller polls the SADP status and analyses the status of the query to conclude whether the SADP executes the completed commands or there is an error, please see OEM protocol for details.

• CAN Protocol

This communication protocol is used for local area network CAN communication. The SADP and parameters are controlled by reading and writing registers without checking the SADP status, and the SADP will automatically upload the current status when the status are changed, please see CAN protocol for details.

4.2 DT Protocol Format

Whenever SADP receives a string, it will parse the string and verify whether the address matches and the syntax is correct, and return the status of the first command execution. If there are multiple commands, sending commands with empty data in the data area can query the status. 9

Data format

Sequence number	Function	Number of bytes	Description
1-2	Address	2	The target communication address, each drive on CAN bus should be set to a unique address number, value ranges from 0-255. It will communicate properly only if the addresses match, otherwise it will ignore the received commands
3	Direction	1	> indicates sending data
4+n	Data	n	ASCII command string, please see operation commands for details
5+n	Terminator	1	The carriage return string 0x0D, every communication data should end with a carriage return

DT send commands protocol format

Sequence number	Function	Number of bytes	Description
1-2	Address	2	The source device address, which matches the sending address, indicates which address returned the data.
3	Direction	1	< indicates return data
	Status	2	See status table
	Special Symbols	1	':' this mark is not displayed if no return data
4+n	Data	n	Return data string, please see the corresponding return data in the operation commands for details
5+n	Terminator	1	The carriage return string 0x0D, every communication data should end with a carriage return

DT return data protocol format

4.3 OEM Protocol Format

The communication protocol includes communication sequences and check bit, which can effectively prevent data loss, so it is recommended to use OEM protocol to communicate with SADP. During operation, the controller queries the SADP status by sending an empty command and analyzes the queried status to determine whether the SADP executes a completed command or an error occurs, sending commands with empty data in the data area can query the status.

Send commands format

Data Fields	Data Type	Number of bytes	Description
Frame head	UInt8	1	Fixed value 0XAA, indicating the start of data
Address	UInt8	1	The communication address, each drive on each bus should be set to a unique address number, value range is from 0-0xff. It will communicate properly only if the addresses match, otherwise it will ignore the received commands
Length of command string	UInt8	1	Data length of the data area
Command string	Char	n	ASCII command string, please see operation commands for details
Checksum	UInt8	1	The 8-bit checksum is calculated from the header of the frame to the end byte of the data area. The value obtained takes the last 8 bits of data

OEM send commands protocol format

Return data format

Data fields	Data type	Number of bytes	Description
Frame head	UInt8	1	Fixed value 0X55, indicating the start of data
Address	UInt8	1	Communication address, each drive on each bus should be set to a unique address number, the address is the original address of the device when returning data
Status	UInt8	1	Current module status, see status table
Length of return string	UInt8	1	Data length of the data area, when the data area has no data, the data length is 0
Return string	Char	n	ASCII code return data, see operation commands for details
Checksum	UInt8	1	The 8-bit checksum, which is calculated from the header of the frame to the end byte of the data area. The value obtained takes the last 8 bits of data

OEM return data protocol format

4.4 CAN Protocol Format

Sending data format

Frame type: data frame

Message identifier type: extended frame

ID: Use the extended ID, see the following table

DLC: Data length is 8

Data field: The data length is fixed 8 bytes, containing communication sequences, register address, and data as shown in the table below.

ID data format

bit23-bit16	bit15-bit8	bit7-bit0
Commands	Source device address	Target device address

Data area data format

byte7	byte6-byte5	byte4	byte3-byte0
Communication sequences	Object index	Object sub-index	4 bytes of data

CAN commands protocol format

Commands: Used to indicate the type of operation to change the frame data, as shown in the following table.

Commands	Function	Description
0x0000	Resnpgondi	Read and write data return
0x0001	Write	Writing object dictionary Return value: status, see status table
0x0002	Read	Read the object dictionary, if there is no corresponding object dictionary, then no data is returned.
0x0003	Process Data	Used for uploading real time data which does not require an answer, e.g. status change, automatically upload via this command
0x0004	Heartbeat	The heartbeat data uploaded at regular intervals, which can be used to detect whether the device is online or not, the heartbeat uploaded data is the node status, the status information is in the status table
0x0080	Alarm	The device will automatically report the commands if error occurs, the error information is shown in the status table

Original address: Sending end address

Destination address: Receiving end address

Communication sequences: It is used to distinguish which data frame is sent and received, and it is recommended that the user + 1 the communication sequences before sending each data frame, so that each data frame is different.

Object index: The index object is 16-bit index address data used to define the object dictionary table, the functions of the object dictionary are described in object dictionary Table. For detailed information, see the CAN object dictionary chapter.

Object sub-index: sub-index is 8-bit sub-index address data, used together with the object index to define the object dictionary table.

Data: The sub-index is an 8-bit sub-index address data, which is used together with the object index to define the object dictionary table.

4.5 Commands Parameter Range Factor K

The current pump has four dispense volume: 100 μL , 250 μL , 500 μL and 1000 μL ; the command parameter range K is 10 for the 100 μL displacement pump, 4 for the 250 μL displacement pump, 2 for the 500 μL displacement pump and 1 for the 1000 μL displacement pump.

5. Serial Interface Commands

This chapter describes the data format in partial data area of the OEM, DT protocol, i.e. the format of the operation commands. The data is an ASCII string and multiple commands sets can be sent at the same time, and SADP will parse and execute the command sets one by one. These commands are classified by function as follows.

- ◆ Initialization commands
- ◆ Control commands
- ◆ Parameter read/write commands
- ◆ Query commands
- ◆ System control commands

5.1 Commands Syntax

The main format of the SADP send commands are as follows:

<CMD><n1,n2,n3><CMD><n1,n2,n3><CMD><n1,n2,n3><CMD><n1,n2,n3>

SADP returns data in ASCII format as follows.

<n1,n2,n3>

Among them:
<CMD>: indicates commands by the letters a-z and A-Z, up to two letters, see commands details
<n1,n2,n3>: indicates the command parameters, multiple parameters are separated by ' , ', and the command without parameters can be empty. If you need part of the parameters by default, you can fill in the corresponding semicolon before the parameters, for example, the second parameter will be an empty command ID1000,,2. If the next part of the parameters are empty and can be omitted, for example, the last two parameters will be empty, command ID1000 means the last two parameters are empty.
Caution:
1) <> is used to differentiate data blocks and does not need to be sent.
2) Commands are case-sensitive.
3) The command letters should be maximum of 2 letters, and the commands of two letters should be an uppercase letter following a lowercase letter, and the commands of one letter should be expressed in uppercase, the special string ? is the query command, {} is the cycle control command, a separate command with one uppercase letter is the system control commands, and commands with an uppercase letter plus a lowercase letter is the run control commands.

5.2 Status

Each command has return status, which indicates whether the commands were executed successfully, whether the device is error, busy, idle, etc. The status is 1 byte hexadecimal data.

Our convention.

0-9: Working status
10-19: Command executing error status
>=20: Equipment failure.

Value	Function	Description
0	Free	Device idle status
1	Busy	The device is busy and does not accept the commands
2	Executed successfully	Commands executed successfully
3	Execution complete	Command execution completed
4	Liquid level detected	Serial communication mode, when detecting liquid level command [n1]=1 turns on the automatic report function of detecting liquid level, the status is automatically uploaded after detecting liquid level, and other modes obtain whether liquid level is detected by querying register 2
...		
10	Parameter exceeds limit	Command parameters out of range
11	Parameter error	Command parameters error
12	Syntax errors	Command syntax error
13	Invalid commands	Commands does not support
14	Wrong address	Read and write parameter address error
15	Write inhibit	This address is inhibited from being written in
16	Read inhibit	This address is inhibited from being read out
17	Uninitialized	Uninitialized
...		
20	Disposable tips not detected alarm	Allow aspiration and dispense of liquid after alarm
21	Eject disposable tip failure alarm	Allow aspiration and dispense of liquid after alarm
22	Timeout alarm	Allow aspiration and dispense of liquid after alarm
23	Aspiration of clot detection alarm	Allow aspiration and dispense of liquid after alarm
25	Aspiration of air detection alarm	Allow aspiration and dispense of liquid after alarm
...		
50	Motor blockage error	Prohibit the aspiration and dispense of liquid, need to re-initialize and troubleshoot
51	Drive failure	Prohibit the aspiration and dispense of liquid, need to re-initialize and troubleshoot
52	Optical coupler error	Prohibit the aspiration and dispense of liquid, need to re-initialize and troubleshoot
53	Optical coupler of disposable tips error	Prohibit the aspiration and dispense of liquid, need to re-initialize and troubleshoot
54	Sensor error	Prohibit the aspiration and dispense of liquid, need to re-initialize and troubleshoot
55	Storage error	Prohibit the aspiration and dispense of liquid, need to re-initialize and troubleshoot

5.3 Commands Details

This section describes the detailed individual command, and the mounting functions are divided into the following categories.

◆ Initialization commands

◆ Control commands

◆ Parameter setting commands

◆ Query commands

◆ Circulation control commands

Caution:

[] indicates optional parameters, if the optional parameters are empty it will be executed according to the default parameter in the protocol.

The content inside <> is a command, the mark <> does not need to be sent, it is just used to distinguish between letters and commands.

Initialization commands

<It>[n1], [n2], [n3] Initialize the device to the initial position

Used to initialize the SADP, automatically control the SADP piston to position 0, and automatically eject the disposable tips according to the parameters.

Command value	Parameters	Data range	Unit	Default value	Description
It	[n1]	200~64000	ustep/s	16000	Velocity during initialization
	[n2]	0~100	%	100	The power of the initialization process. Percentage of maximum power
	[n3]	0	None	0	Disposable tips will be ejected regardless of whether the disposable tips is detected or not
		1			Disposable tips detected, will eject disposable tips
		2			No need to eject the disposable tips

Return data: See the status table.

SADP control commands

<Ia>n1, [n2], [n3] Aspirate

Used for aspiration, automatically control the SADP piston to 0 position, and automatically eject the disposable tips according to the parameters, please note that the initialization process must ensure that the SADP in the appropriate position, otherwise may be damaged because the SADP in the initialization process touch something else.

Command value	Parameters	Data range	Unit	Default value	Description
Ia	n1	0~100000/K	0.01ul	0	Aspirate volume
	[n2]	0~2000/K	1ul/s	500/K	Aspirate velocity
	[n3]	0~2000/K	1ul/s	10	Cut off velocity

[n1] Aspirate volume:

The parameter is required, theoretically speaking, the aspirate volume is the volume of aspiration to the disposable tips, but due to different physical factors, the aspirate volume and the target volume are different, for high precision aspiration, they need to combine the aspirate velocity, liquid viscosity, aspirate volume, disposable tips type and other physical characteristics to do the corresponding calibration curve.

[n2] Aspirate rate:

The flow should be set according to the physical properties of the liquid. For example, viscosity and surface tension. Too fast a flow for different liquids may result in low pressure in the SADP chamber, and may also result in continued liquid flow into the disposable tips if the plunger stops, which can lead to mis-aspiration. A reasonable approach is to set the liquid flow rate to match the aspirate rate.

[n3] Cut off velocity:

This parameter is used to control the velocity at which the fluid finally stops and is set according to the different physical properties of the fluid.

Return data: see status table.

<Da>n1, [n2], [n3], [n4] Dispense

For dispense, automatic control of the SADP piston to position 0 and automatic eject the disposable tips depending on the parameters.

Command value	Parameters	Data range	Unit	Default value	Description
Da	n1	0~100000/K	0.01ul	0	Dispensed volume
	[n2]	0~10000			Re-aspirate volume
	[n3]	0~2000/K	1ul/s	500/K	Dispense velocity
	[n4]			10	Stop velocity

[n1] Aspirate volume:

The parameter is required, theoretically speaking, the aspirate volume is the volume of aspiration to the disposable tips, but due to different physical factors, so the aspirate volume and the target volume are different, for high precision aspiration, they need to combine the aspirate velocity, liquid viscosity, aspirate volume, disposable tips type and other physical characteristics to do the corresponding calibration curve.

[n2] Re-aspirate volume:

In order to avoid the droplet of the tips affecting the accuracy and causing cross-contamination, we design the function of automatically re-aspirating specified volume of liquid after the dispense is completed, which can well avoid the droplets problem. The re-aspirate volume is affected by the dispense volume, disposable tip type, reagent viscosity, etc. The dispense accuracy is affected by the dispense volume, for different dispense volumes, users need to debug and change the parameters by themselves to achieve anti-droplets and ideal dispense accuracy.

[n3] Dispense velocity:

The flow should be set according to the physical properties of the liquid, such as viscosity and surface tension. For different liquids, flow too fast may lead to high pressure in the SADP chamber, and a reasonable method is to set the liquid flow to match the dispense rate.

[n4] Stop velocity:

It is used to set the velocity at the moment of dispense completion, decelerate to stop velocity after the final stage of dispense completion and then stop, note that the stop velocity must be less than or equal to the dispense velocity. Higher stop velocity can quickly cutoff the dispensed fluid, which has some improvement on droplets of the tips.

Return data: see status table.

<Dt>[n1],[n2] Eject disposable tip

Used to control the internal motor of the module to eject the disposable tips. The commands are sent to determine if eject the disposable tips by querying the status of disposable tips in place flag.

Command value	Parameters	Data range	Unit	Default value	Description
Dt	n1	200~32000	ustep/s	16000	Velocity during eject disposable tips
	[n2]	0	None	0	Disposable tips will be ejected regardless of whether the disposable tips is detected or not
		1			Disposable tips detected, will eject disposable tips

Return data: see status table.

<Ld>[n1],[n2] Liquid level detection

It is used to detect the reagent surface. When liquid level detection, drop the SADP with disposable tips 5mm spacing with reagent vessel firstly, then send the liquid level detection commands and continue to drop, when the disposable tips touch the liquid level, the SADP will send the signal of liquid level detection; users can view the signal of liquid level detection in three ways: monitoring communication data, monitoring digital output and automatically querying register status.

Command value	Parameters	Data range	Unit	Default value	Description
Ld	n1	0~1	None	1	0: The liquid level is not automatically reported after it is detected, and it is necessary to confirm whether the liquid level is detected by querying whether the status detection level flag is 1. 1: Automatically report the status after detecting the liquid level, the status of the detected liquid level flag is 1
	[n2]		ms	10000	0: No timeout detection. Other values: timeout period, timeout period without detecting atmospheric pressure change will report timeout error

Return data: see status table.

<Mp>n1,[n2] Move to absolute position

Control piston moves to the absolute position.

Command value	Parameters	Data range	Unit	Default value	Description
Mp	n1	0~maximum stroke	ustep	0	Location
	[n2]	0~64000	ustep	8000	Operating speed
	[n3]	0~32000	ustep	160	Stop velocity

Return data: see status table.

<Up>n1,[n2] Move up to relative position

Control the piston move up to the specified position of aspiration relative to the current position.

Command value	Parameters	Data range	Unit	Default value	Description
Up	n1	0~32000	ustep	0	Relative position
	[n2]	0~64000	ustep	8000	Operating speed
	[n3]	0~32000	ustep	160	Stop velocity

Return data: see status table.

<Dp>n1,[n2],[n3],[n4] Move down to relative position

Control the piston move down to the specified position of dispensed liquid relative to the current position.

Command value	Parameters	Data range	Unit	Default value	Description
Dp	n1	0~32000	ustep	0	Relative position
	[n2]	0~64000	ustep	8000	Operating speed
	[n3]	0~32000	ustep	160	Stop velocity
	[n4]	0~32000	ustep	0	Rebound distance

Return data: see status table.

[n4] Rebound distance:

This parameter is optional and enables a quick rebound of the piston to avoid droplets and specific volume re-aspiration.

Parameter setting

Register

The registers are used for user setting and viewing the parameters of SADP for flexible application.

Register address	Reading and writing	Data range	Unit	Default value	Description
1	R/W			0	Current status (see status table), write 0 to clear the error, return value is the same as the query status commands ?
2	R	0-1		0	Whether the liquid level is detected

3	R	0-1	None	0	Whether there is a disposable tip; 0: no disposable tip, 1: have disposable tip
4	R				Current pressure ADC value
...					
6	R				Disposable tips model (reserved register)
7	R	0-256		0	Query the current command pointer address, and execute the command sets from 0 where the string is executed
8	R			0	AUXIN value
9	R			0	AUXOUT value
10	R			0	AUXOUT output configuration. 0: Output 10 ms high level voltage after detecting liquid level 1: Output high level voltage when liquid level is detected 2: Output low level voltage when liquid level is detected
...					
20	R	0-32000	ustep		Current motor position
21	R	0-32000	ustep/s		Current motor velocity
22	R	0-1000/k	ul/s		Current fluid velocity
23	R	0-64000	ustep/s	16000	Maximum motor velocity
24	R	0-32000	ustep/s	160	Minimum motor velocity
25	R	0-320000	ustep/ss	240000	Motor acceleration
26	R	0-320000	ustep/ss	240000	Motor deceleration
27	R	0~100	%	100	Motor rated power
28	R		ustep		Maximum stroke
29	R	1000/K	ul		Maximum volume
30	R				Pressure sensor nulling (reserved register)
31	R		ul		Disposable tip volume (reserved register)
32	R	0~3200	ustep		Backlash
33	R	0~2300	mA	500	Rated current
36	R	0-1			0: Disable pressure sensor. 1: Enable pressure sensor.
...					
35	R	0~1000	ul		Current DiTis reagent volume, this value is a rough calculation of the reagent volume based on the current pressure and should only be used as a reference. (reserved register)
...					

41	R/W	0-64000	ustep/s	0	Velocity of eject disposable tips
42	R/W	0-64000	ustep		Distance of eject disposable tips
43	R/W	0-3		0	HEX code, corresponding bit 1 to enable this function Bit0: detect whether there is a disposable tips before aspirate and dispense liquid, no disposable tips error is reported Bit1: Not using DiTis detection optocoupler
...					
54	R/W	1-100		10	liquid level detection coefficient, the smaller the value the more sensitive
...					
60	R/W	0-0x07		0	Pressure abnormality detection enable, HEX code, corresponding bit is 1 to enable the function Bit0: Aspiration of clot detection enable Bit2: Aspiration of air detection enable
...					
70	R/W	0-100		10	Aspiration of clot detection coefficient, the smaller the value the more sensitive
72	R/W	0-100		20	Aspiration of air detection coefficient, the larger the value the more sensitive
...					
80	R/W			38400	Serial interface baud rate
81	R/W			500	CAN baud rate
82	R/W	0-1		0	Active data upload after motion completed 0: Don't upload. 1: Upload
83	R/W	0-10000	ms	1000	Heartbeat interval time 0: No heartbeat data Other values: heartbeat interval time, in ms
84	R/W	0-255		1	Device address
...					
90	R				Software version
91	R				Device model, which is the same for each SADP
92	R				Device sequence number, unique value of each SADP
...					
100	R/W	0-50000	um	0	Z-axis speed for level detection, automatic control of the Z-axis during liquid level detection, no movement of the Z-axis when set to 0;
101	R/W	0-180000	um		The position of the Z-axis when the tip of the TIP head is moved to the bottom of the test tube, the level detection or the travel limit when following the level of the aspirate and dispense liquid;
102	R/W	0-180000	um		Position of the Z-axis when the tip of the TIP head moves to the mouth of the test tube and the stroke limit when the dispense level follows;
103	R/W	0-180000	um		The tip of the TIP head is moved to the position of the Z-axis at the test diameter, the lower half of the tube is conical and is used when the aspirate and dispense levels follow;
104	R/W	0-180000	mm ²		The cross-sectional area of the inner chamber of the test tube, the liquid level follows when aspirating and dispensing, when set to 0 the aspirating and dispensing Z-axis does not move

<Wr>n1,n2 Write register

Write register.

Instruction value	Parameters	Data range	Unit	Default value	Description
Wr	n1	0~100	None	None	Register address
	n2				Write data

<Rr>n1,[n2] Read register

Read registers, read the specified number of registers from the starting address, read registers larger than 1, return data separated by ','.

Instruction value	Parameters	Data range	Unit	Default value	Description
Rr	n1	0~100	None	None	Start address
	n2				Number of registers

Return data: the status part is shown in the status table, the data area part is the data to be fetched, and the return data of multiple addresses are separated by ','.

System operation commands

Status Search

<?>

Send commands? to query status.

Return data: see status table

Circulation control commands

The circulation control commands are used to control the cycle of the command string, the cycle can be nested, a maximum of 20 cycles including nested cycles are supported in an command string.

{ Cycle start position

Command value	Parameters	Data range	Unit	Default Value	Description
{	None		None	0	Cycles start

}[n1] End-of-cycle position

Command value	Parameters	Data range	Unit	Default Value	Description
}	[n1]	0-2147483647		0	No parameter or 0: Endless cycle Other values: Cycle times

{: Cycle start

}: The end of the cycle, followed by the number of cycles, if there is no number of cycles or the data is empty, it means endless cycle.

Note: Cycles can be nested

<L>n1 Delay

The delay is used for internal system delay, mainly as a delay between two command executions.

Command value	Parameters	Data range	Unit	Default Value	Description
L	n1	0~2147483647	ms	None	Delay time

Return data: see status table.

<T> Halt command

Halts the currently executing commands.

Command value	Parameters	Data range	Unit	Default Value	Description
T	None				

<U> Reset command

The commands are used as a device restart reset operation.

Command value	Parameters	Data range	Unit	Default Value	Description
U	None			123456	Only the parameter be 123456, the reset will be successful.

<M>n1 Restore factory settings command

After executing the commands, you need to restart the device to take effect.

Command value	Parameters	Data range	Unit	Default Value	Description
M	n1			123456	The parameter must be 123456 in order to restore the factory settings successfully

<S> Power down to save setting parameters

After executing the commands, the modified register parameters will be saved by power down.

6. CAN Object Dictionary

The object dictionary includes 8-bit primary index and 8-bit sub-index, different operations and parameter read/write are indicated by ordered index numbers, all data including read and write permissions. The control commands include initialization, aspirate, dispense, etc., all follow different primary indexes to indicate different commands, and sub-indexes are different parameters, the number of parameters is the same as the number of serial command parameters, for example, the serial interface commands It[n1][n2][n3], for instance, need to modify the default parameters before executing the commands.

6.1 Control commands

Function	Primary index	Sub-index	Read and write permissions	Data range	Default value	Function
Initialization	0x4000	0	W	200-64000 (ustep/s)	Required	Initialization, velocity during initialization of parameters
		1	RW	0-100(%)	100	Power during initialization
		2	RW	0-2	0	Eject disposable tip during initialization 0: Disposable tips will be ejected regardless of whether the disposable tips is detected or not 1: Disposable tips is detected and will eject 2: No need to eject the disposable tips
Aspirate	0x4001	0	W	0-100000 (0.01ul)/K	Required	Aspiration volume, resolution is 0.01ul, for example, parameter is 10000 if aspirated 100 μ L
		1	RW	0-2000/K	500/K	Aspiration velocity, in μ L/s
		2	RW	0-2000/K	10	Cut off velocity, in μ L/s

Dispense	0x4002	0	W	0-100000 (0.01ul)/K	Required	Dispense volume, resolution is 0.01ul, for example, parameter is 1000 if aspirated 10μL
		1	RW	0-100000 (0.01ul)/K	0	Re-aspirate volume, resolution is 0.01ul, for example, parameter is 200 if aspirated 2 μL
		2	RW	0-2000/K	500/K	Aspirate velocity, in μL/s
		3	RW	0-2000/K	10	Cut off velocity, in μL/s
Moveto absolute position	0x4003	0	W	0-32000	Required	Position value, in ustep
		1	RW	0-64000	16000	Operating speed, in ustep/s
		2	RW	0-32000	160	Stop velocity, in ustep/s
Relative upward motion	0x4004	0	W	0-32000	Required	Motion distance, in ustep
		1	RW	0-64000	16000	Operating speed, in ustep/s
		2	RW	0-32000	160	Stop velocity, in ustep/s
Relative downward motion	0x4005	0	W	0-32000	Required	Motion distance, in ustep
		1	RW	0-64000	16000	Operating speed, in ustep/s
		2	RW	0-32000	160	Stop velocity, in ustep/s
		3	RW	0-3200	0	Re-aspirate value, in ustep
Eject disposable tip	0x4006	0	W	200-32000 (ustep/s)	Required	Velocity during ejecting the disposable tips process , in ustep/s
		1	RW	0-1	0	Eject disposable tips model 0: Disposable tips will be ejected regardless of whether the disposable tips are detected or not 1: Disposable tips is detected and will eject
Liquid level detection	0x4007	0	W	0-1	Required	Data reporting type 0: Didn't automatically reported after liquid level detected, and it is necessary to confirm whether the liquid level is detected by querying whether the status detection level flag is 1. 1: Automatically report the status after detecting the liquid level, the status of the detected liquid level flag is 1
		1	RW	0-20000 (ms)	0	Timeout detection time 0: No timeout detection. Other values: Timeout period, timeout period without detecting atmospheric pressure change will report timeout error
Stop	0x4008	0	W	0	0	Stop motion and liquid level detection
Power down save	0x5000	0	W	0	0	Power down to save data
Reset reboot	0x9F00	3	W	0	0	Reset reboot

6.2 Register reading and writing

We agree that the primary index 0x2000 is the SADP internal register, the sub-index register address and the register address are shown in 2.3.3 register table

6.3 Process data

The SADP automatically uploads process data via commands 0x03, and the process data dictionary is as follows

Function	Primary index	Sub-index	Read and write permissions	Data range	Default value	Function
Liquid level detected	0x7000	0	R	1	1	Automatic uploading of this status value upon detection of the liquid level1
Disposable tips detection status	0x7001	0	R	0-1	0	Disposable tips detection status Automatic upload of the disposable tip status when it changes 0: No disposable tips 1: Have disposable tips
Completion of Motion	0x7002	0	R	0-255	0	Data uploaded by the campaign completion initiative 0: Normal Other values: error status, see status table

6.4 Heart rate data

The SADP sends heartbeat data via commands 0x04, which the master can use to detect if the device is online

6.5 Alarm data

SADP sends an alarm message through the commands 0x80, the alarm data formats are as follows, same as the error message of the query status

Code	Description	Code	Description	Code	Description
20	Disposable tips not detected	50	Motor blockage error		
21	Eject the disposable tips failure	51	Drive failure		
22	Timeout error	52	Zero position optocoupler error		
23	Aspiration of clot detection alarm	53	Disposable tips optocoupler error		
25	Aspiration of air detection alarm	54	Sensor error		
...		55	Storage error		

7. Application Process

Note that the serial communication DT protocols have no data checksum, so it is best to use only for serial interface debugging, the device should still use OEM protocol, the data formats of OEM protocol and DT protocol in data area are the same, only the frame head and frame tail data formats are different. When OEM protocol is applied, each operation should check the status to ensure that the commands are executed before the next operation is performed. The SADP address of the following data is 1.

7.1 Example of DT Protocol Applications (String Mode Sending)

Initialization

	Data	Description
Send	1>It16000,100,0	1 target device address; 16000 initial operating speed is 16000 microsteps/sec; 100 initial operating power is 100%; 0 In initialization process, disposable tips will eject regardless of whether the tips are detected or not
Receiving	1<2	1 target device address; 2 command executed successfully status (see status table for others)

Liquid level detection

	Data	Description
Send	1>Ld1,5000	1 target device address; 1 automatically report status after detecting the liquid level; 5000 detection timeout period is 5000 milliseconds
Receiving	1<2	1 target device address; 2 command executed successfully status (see status table for others)
Receiving	1<4	1 target device address; 4 liquid level status detected; while the blue LED on the PCB is always on

Aspirate

	Data	Description
Send	1>la10000,200,10	1 target device address; 10,000 aspirate 100 microliters; 200 operating speed is 200 microliters/sec during aspiration; 10 aspirate cutoff velocity is 10 microliters/sec
Receiving	1<2	1 target device address; 2 command executed successfully status (see status table for others)

Dispense

	Data	Description
Send	1>Da1000,500,200,100	1 target device address; 1000 dispense 10 microliters; 500 re-aspirate volume 5 microliters; 200 dispense process operating speed is 200 microliters/sec; 100 dispense cutoff velocity 10 microliters/sec
Receiving	1<2	1 target device address; 2 command executed successfully status (see status table for others)

Query current status

	Data	Description
Send	1>?	1 target device address; used to query the current SADP status, equivalent to reading the register address 1 set in 2.3.3 parameter
Receiving	1<0	1 target device address; 0 idle state (see status table for others)

Read parameters

	Data	Description
Send	1>Rr3	1 target device address; 3 check whether there is a disposable tip (other register addresses to see 2.3.3 parameter settings)
Receiving	1<2:0	1 target device address; 2 command executed successfully (see status table for others); 0 no disposable tip

Setting parameters

	Data	Description
Send	1>Wr54,10	1 target device address; 54 set the liquid level detection coefficient (other register addresses see 2.3.3 parameter settings); 10 set liquid level detection coefficient to 10
Receiving	1<2	1 target device address; 2 command executed successfully (see status table for others)

7.2 Example of OEM Protocol Applications (HEX Mode Sending)

Description: The OEM protocol is an encapsulation of the DT protocol, and the command string and return string are described in the DT protocol.

Initialization

	Data	Description
Send	AA 01 0D 49 74 31 36 30 30 30 2C 31 30 30 30 2C 30 85	AA frame head; 01 target device address; 0D command string length; 49 74 31 36 30 30 30 30 2C 31 30 30 30 2C 30 command string "lt16000,100,0"; 85 frame checksum
Receiving	55 01 02 00 58	55 frame head; 01 target device address; 02 command executed successfully status (see status table for others); 00 return string length; 58 frame checksum

Liquid level detection

	Data	Description
Send	AA 01 08 4C 64 31 2C 35 30 30 30 85	AA frame head; 01 target device address; 08 command string length; 4C 64 31 2C 35 30 30 30 30 command string "Ld1,5000"; 85 frame checksum
Receiving	55 01 02 00 58	55 frame head; 01 target device address; 02 command executed successfully status (see status table for others); 00 return string length; 58 frame checksum

Aspirate

	Data	Description
Send	AA 01 0E 49 61 31 30 30 30 30 2C 32 30 30 2C 31 30 9F	AA frame head; 01 target device address; 0E command string length; 49 61 31 30 30 30 30 30 2C 32 30 30 2C 31 30 command string "la10000,200,10"; 9F frame checksum
Receiving	55 01 02 00 58	55 frame head; 01 target device address; 02 command executed successfully status (see status table for others); 00 return string length; 58 frame checksum

Dispense

	Data	Description
Send	AA 01 12 44 61 31 30 30 30 2C 35 30 30 2C 32 30 30 2C 31 30 30 5F	AA frame head; 01 target device address; 12 command string length; 44 61 31 30 30 30 2C 35 30 30 2C 32 30 30 2C 31 30 30 30 command string "Da1000,500,200,100"; 5F frame checksum
Receiving	55 01 02 00 58	55 frame head; 01 target device address; 02 command executed successfully status (see status table for others); 00 return string length; 58 frame checksum

Query current status

	Data	Description
Send	AA 01 01 3F EB	AA frame head; 01 target device address; 01 command string length; 3F command string "? "; EB frame checksum
Receiving	55 01 00 00 56	55 frame head; 01 target device address; 00 idle status (see status table for others); 00 return string length; 58 frame checksum

Read parameters

	Data	Description
Send	AA 01 03 52 72 33 A5	AA frame head; 01 target device address; 03 command string length; 52 72 33 command string "Rr3"; A5 frame checksum
Receiving	55 01 02 01 30 89	55 frame head; 01 target device address; 02 command executed successfully (see status table for others); 01 return string length; 30 return string "0"; 58 frame checksum

Setting parameters

	Data	Description
Send	AA 01 07 57 72 35 34 2C 31 30 71	AA frame head; 01 target device address; 07 command string length; 57 72 35 34 2C 31 30 command string "Wr54,10"; 71 frame checksum
Receiving	55 01 02 00 58	55 frame head; 01 target device address; 00 idle status (see status table for others); 00 return string length; 58 frame checksum

7.3 Examples of CAN Protocol Applications

Description: View 6.1 control commands, each with a unique primary index and multiple sub-indexes, sending frame data with non-zero sub-indexes first, and finally sending frame data with zero sub-indexes, SADP starts to move when it receives commands with zero sub-indexes.

Initialization

	ID	Data	Description
Send first	00 01 00 01	01 40 00 01 00 00 00 64	00 01 write commands; 00 source device address; 01 target device address; 01 communication sequences; 40 00 primary index (initialization control); 01 sub-index (initialization process power setting); 00 00 00 64 data (set initialization process power to 100%)
Receiving	00 00 01 00	01 40 00 01 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 01 communication sequence (consistent with the sequence in the received commands); 40 00 primary index; 01 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)

Resend	00 01 00 01	02 40 00 02 00 00 00 00	00 01 write commands; 00 source device address; 01 target device address; 02 communication sequences; 40 00 primary index (initialization control); 02 sub-index (initialization process to disposable tip mode setting); 00 00 00 00 data (eject the disposable tip regardless of whether it is detected or not during initialization process)
Receiving	00 00 01 00	02 40 00 02 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 02 communication sequence (consistent with the sequence in the received commands); 40 00 primary index; 02 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)
Last sent	00 01 00 01	03 40 00 00 00 00 3E 80	00 01 write commands; 00 source device address; 01 target device address; 03 communication sequences; 40 00 primary index (initialization control); 00 sub-index (initialization process velocity setting); 00 00 3E 80 data (set initialization process velocity to 16000 microsteps/sec)
Receiving	00 00 01 00	03 40 00 00 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 03 communication sequence (consistent with the sequence in the received commands); 40 00 primary index; 00 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others, see status table for others)

Liquid level detection

	ID	Data	Description
Send first	00 01 00 01	04 40 07 01 00 00 13 88	00 01 write commands; 00 source device address; 01 target device address; 04 communication sequences; 40 07 primary index (liquid level detection); 01 sub-index (liquid level detection timeout setting); 00 00 13 88 data (set liquid level detection timeout to 5000 ms)
Receiving	00 00 01 00	04 40 07 01 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 04 communication sequence (consistent with the sequence in the received commands); 40 07 primary index; 01 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)
Resend	00 01 00 01	05 40 07 00 00 00 00 01	00 01 write commands; 00 source device address; 01 target device address; 05 communication sequences; 40 07 primary index (liquid level detection); 00 sub-index (liquid level detection data reporting type setting); 00 00 00 01 data (automatic report of liquid level detection)
Receiving	00 00 01 00	05 40 07 00 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 05 communication sequence (consistent with the sequence in the received commands); 40 07 primary index; 00 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)
Receive when liquid level is detected	00 03 01 00	E2 70 00 00 00 00 00 04	00 03 process data; 01 address of device sending data; 00 address of device receiving data; E2 communication sequence (consistent with heartbeat sequence, plus one for each transmission); 70 00 primary index (liquid level detected); 00 sub-index; 00 00 00 04 status (liquid level detected)
Receives when a liquid level timeout is detected	00 80 01 00	E9 00 00 00 00 00 00 16	00 80 alarm data; 01 address of device sending data; 00 address of device receiving data; E9 communication sequence (consistent with heartbeat sequence, plus one for each transmission); 00 00 primary index; 00 sub-index; 00 00 00 16 status (timeout error)

Aspirate

	ID	Data	Description
Send first	00 01 00 01	06 40 01 01 00 00 00 C8	00 01 write commands; 00 source device address; 01 target device address; 06 communication sequences; 40 01 primary index (aspirate); 01 sub-index (aspirate velocity setting); 00 00 00 C8 data (aspirate velocity 200 microliters/sec)
Receiving	00 00 01 00	06 40 01 01 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 06 communication sequence (consistent with the sequence in the received commands); 40 01 primary index; 01 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)
Resend	00 01 00 01	07 40 01 02 00 00 00 0A	00 01 write commands; 00 source device address; 01 target device address; 07 communication sequences; 40 01 primary index (aspirate); 02 sub-index (aspiration cutoff velocity setting); 00 00 00 0A data (set aspiration cutoff velocity to 10 microliters/sec)
Receiving	00 00 01 00	07 40 01 02 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 07 communication sequence (consistent with the sequence in the received commands); 40 01 primary index; 02 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)
Last sent	00 01 00 01	08 40 01 00 00 00 00 27 10	00 01 write commands; 00 source device address; 01 target device address; 08 communication sequences; 40 01 primary index (aspirate); 00 sub-index (aspirate volume setting); 00 00 27 10 data (aspirate 100 µl)
Receiving	00 00 01 00	08 40 01 00 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 08 communication sequence (consistent with the sequence in the received commands); 40 01 primary index; 00 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)

Dispense

	ID	Data	Description
Send first	00 01 00 01	09 40 02 01 00 00 01 F4	00 01 write commands; 00 source device address; 01 target device address; 09 communication sequences; 40 02 primary index (dispense); 01 sub-index (re-aspirate volume setting); 00 00 01 F4 data (set re-aspirate volume to 5 µl)
Receiving	00 00 01 00	09 40 02 01 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 09 communication sequence (consistent with the sequence in the received commands); 40 02 primary index; 01 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)
Resend	00 01 00 01	0A 40 02 02 00 00 00 C8	00 01 write commands; 00 source device address; 01 target device address; 0A communication sequences; 40 02 primary index (dispense); 02 sub-index (dispensed velocity setting); 00 00 00 C8 data (set dispense velocity to 200 microliters/sec)
Receiving	00 00 01 00	0A 40 02 02 00 00 00 02	00 00 answer data; 01 answer device address; 00 receive answer data device address; 0A communication sequence (consistent with the sequence in the received commands); 40 01 primary index; 02 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)

Resend	00 01 00 01	0B 40 02 03 00 00 00 64	00 01 write commands; 00 source device address; 01 target device address; 0B communication sequences; 40 02 primary index (dispense); 03 sub-index (dispense cutoff velocity setting); 00 00 00 64 data (set aspiration cutoff velocity to 100 microliters/sec)
Receiving	00 00 01 00	0B 40 02 03 00 00 00 02	00 00 answer data; 01 answer device address; 0B receive answer data device address; 0B communication sequence (consistent with the sequence in the received commands); 40 02 primary index; 03 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)
Last sent	00 01 00 01	0C 40 02 00 00 00 03 E8	00 01 write commands; 00 source device address; 01 target device address; 0C communication sequences; 40 02 primary index (dispense); 00 sub-index (dispense volume setting); 00 00 03 E8 data (dispense 10 µl)
Receiving	00 00 01 00	0C 40 02 00 00 00 00 02	00 00 answer data; 01 answer device address; 0C receive answer data device address; 0C communication sequence (consistent with the sequence in the received commands); 40 02 primary index; 00 sub-index; 00 00 00 02 status (commands executed successfully, see status table for others)

Read parameters

	ID	Data	Description
Send	00 02 00 01	0D 20 00 01 00 00 00 00	00 02 read commands; 00 source device address; 01 target device address; 0D communication sequences; 20 00 primary index (register operation); 01 sub-index (current state, see parameter setting register); 00 00 00 00 data
Receiving	00 00 01 00	0D 20 00 01 00 00 00 00	00 00 answer data; 01 answer device address; 0D receive answer data device address; 0D communication sequence (same sequence as in the received commands); 20 00 primary index; 01 sub-index; 00 00 00 00 data (idle state, see status table for others)
Send	00 02 00 01	0E 20 00 02 00 00 00 00	00 02 read commands; 00 source device address; 01 target device address; 0E communication sequences; 20 00 primary index (register operation); 02 sub-index (whether liquid level is detected, see parameter setting register); 00 00 00 00 data
Receiving	00 00 01 00	0E 20 00 02 00 00 00 01	00 00 answer data; 01 answer device address; 0E receive answer data device address; 0E communication sequence (same sequence as in the received commands); 20 00 primary index; 02 sub-index; 00 00 00 01 data (liquid level detected, if 00 00 00 00 then no liquid level detected)

Setting parameters

	ID	Data	Description
Send	00 01 00 01	0F 20 00 36 00 00 00 0A	00 01 write commands; 00 source device address; 01 target device address; 0F communication sequences; 20 00 primary index (register operation); 36 sub-index (liquid level detection coefficient, see parameter setting register); 00 00 00 0A data (set liquid level detection coefficient to 10)
Receiving	00 00 01 00	0F 20 00 36 00 00 00 02	00 00 answer data; 01 answer device address; 0F receive answer data device address; 0F communication sequence (same sequence as in the received commands); 20 00 primary index; 36 sub-index; 00 00 00 02 data (set up successfully, see status table for others)

8. LED Indicator Functions

There are two LEDs: yellow-green and blue, blue is mainly used to check whether the motor is working, yellow-green light is mainly used to indicate whether the liquid level is detected and whether the disposable tips are detected, by flashing two lights at the same time to indicate the error, as shown in the table below.

LED status	
No color	Free
Blue	Busy status / liquid level detected
Yellow green	Disposable tips detected
Blue light blinking	Errors, the number of blinks corresponds to different errors, see the following table

(LED status)

Number of blinks	Description
1	Disposable tips not detected alarm
2	Eject disposable tips failure alarm
3	Timeout Alarm
4	Aspirate clot alarm
6	Aspiration of air alarm
9	Motor blockage error
10	Drive error
11	Zero position optocoupler error
12	Disposable tips optocoupler error
13	Sensor error
14	Storage error

(Error indication)

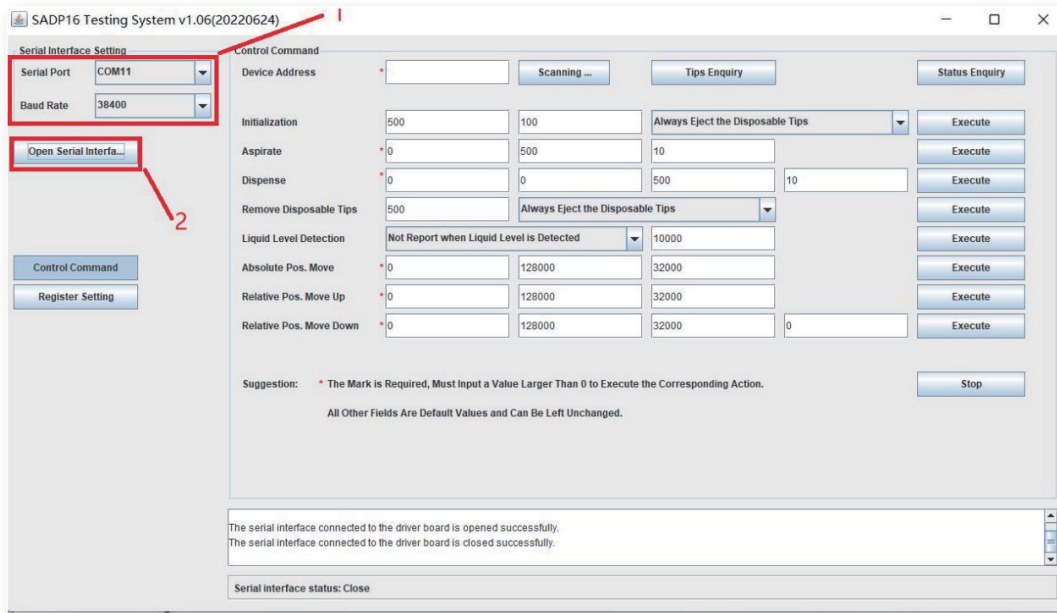
9. Description of Host Computer Test Software

1. Open the SADP20 host computer test software:

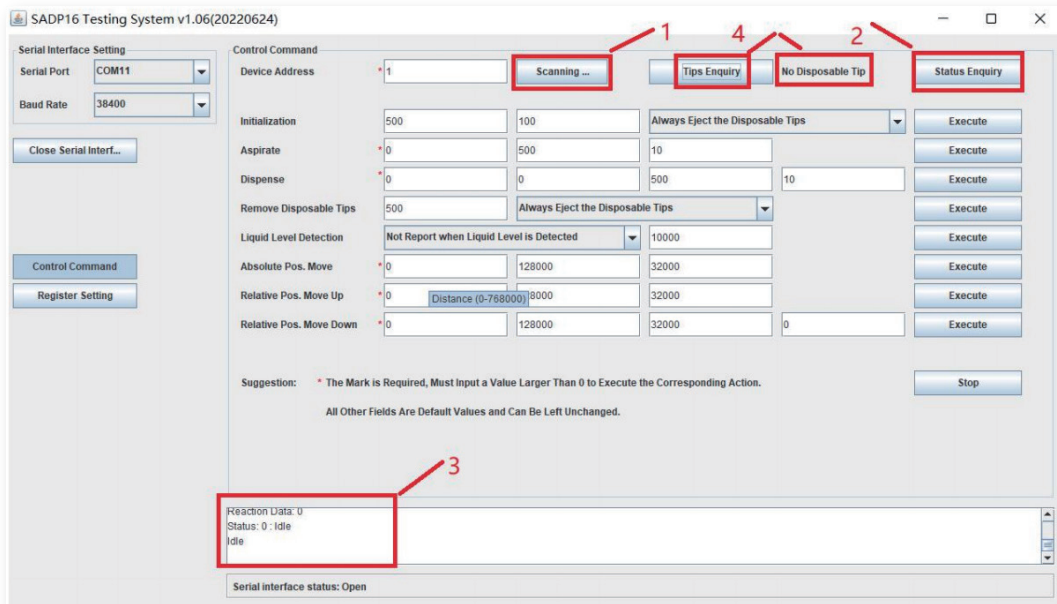
First, the device is connected to the cable and power on, and then open the test software:



2. Serial interface, baud rate selection:
 - A. Select the corresponding serial interface number and the baud rate of 38400 (factory default 38400)
 - B. Click "Open"

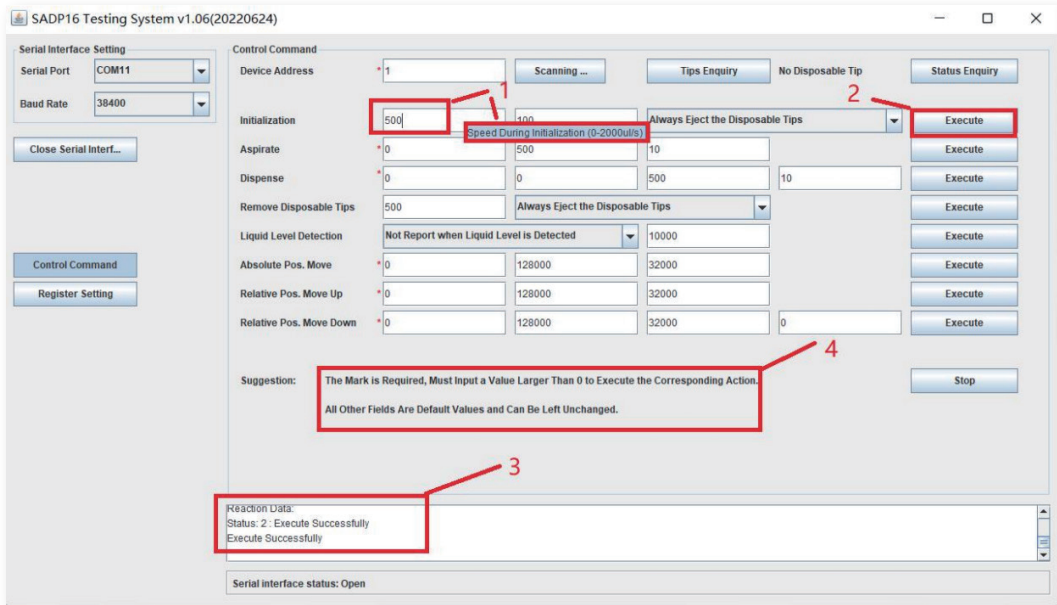


3. Scan device address, query status:
 - A. Click Scan, the default address is 1, the address bar will automatically fill in the input box with the scanned address after the scan is completed, and the address must be scanned before other operations.
 - B. Click on "Status Inquiry"
 - C. Return to current status
 - D. Click "Disposable Tips Query", next to it will show whether the device currently has disposable tips



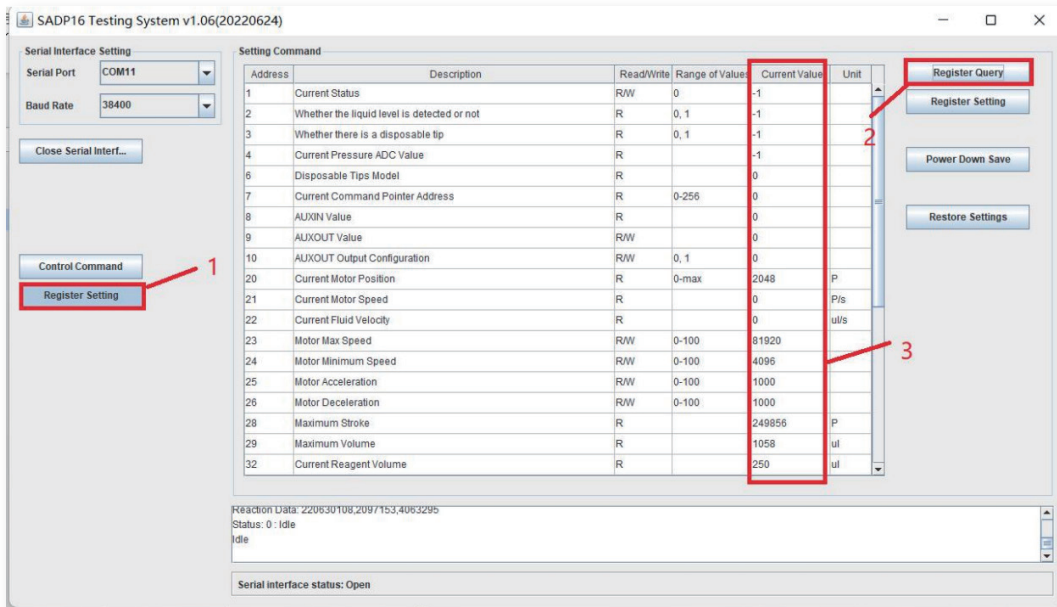
4. Basic operation:

- Mouse over the parameter box (white position of the setting parameter box), stay for about 1 second will display the name, parameter range and unit of the current parameter box.
- Click on "Initialize" then "Execute"
- The command operation returns data, and the execution succeeds in initializing the device normally
- OEM communication data alert



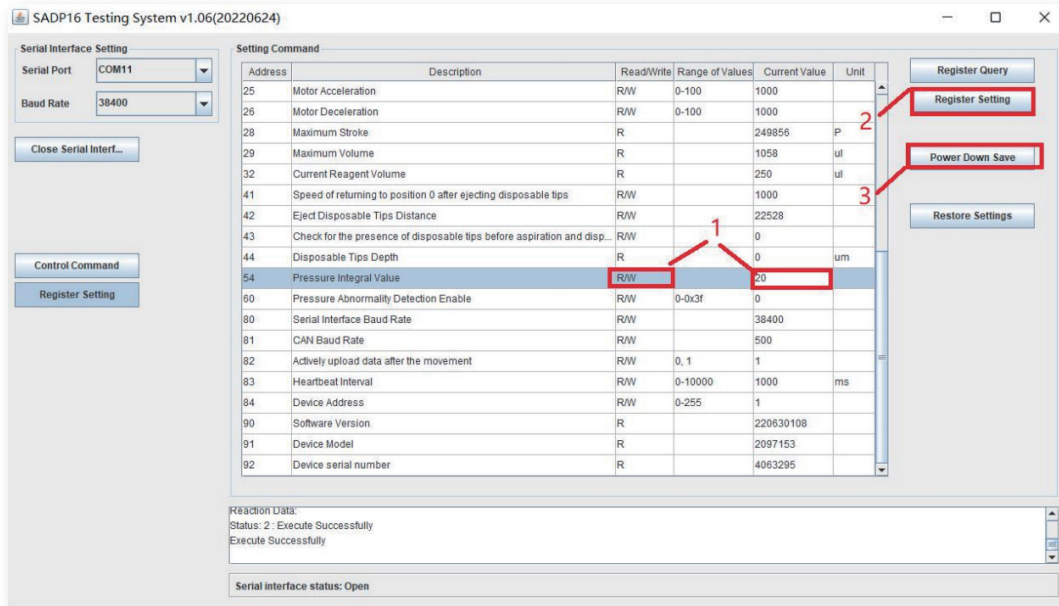
5. Register query:

- Click "Register Setting" to switch the setting command interface
- Click on "Query Register"
- The value of the register is displayed in the "Current Value".



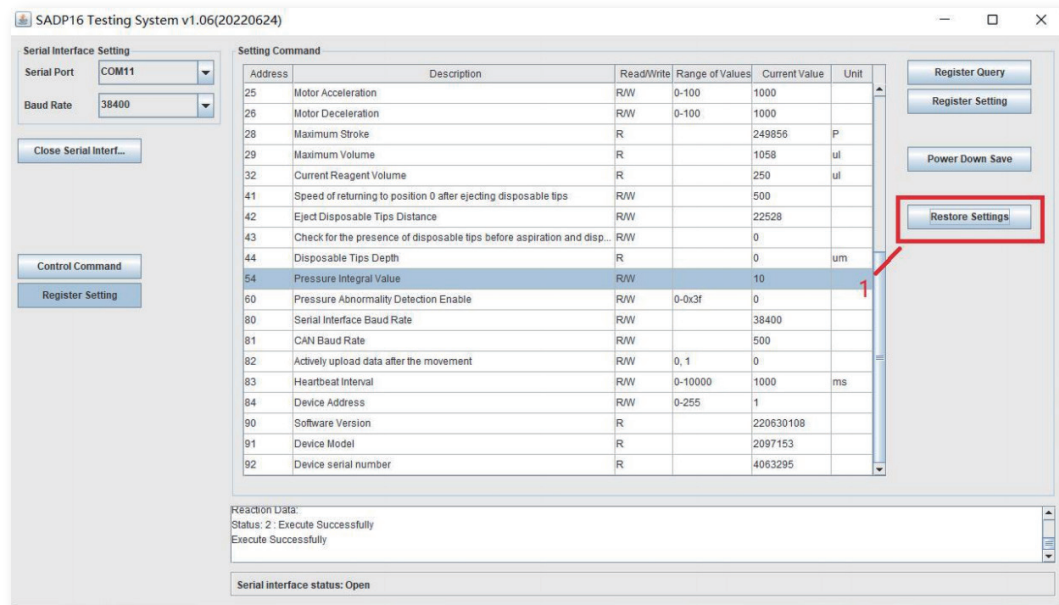
6. Register parameter setting:

- Double-click the location of the "Current Value" of the register parameter that needs to be modified (only if it can be written), fill in the modified value, press Enter or click on the blank area
- Click on "Register Settings"
- Click on "Power Down Save"
- Device power off and restart (after the device power off and restart, the host computer software needs to operate step 2 and switch the serial interface once)



7. Restore factory settings

- Click "Restore Factory Settings", the device power off and restart (after the device power off and restart, the host computer software needs to operate step 2, switch the serial interface once)



10. Atmospheric Pressure Testing

SADP has pressure sensor that can monitor the atmospheric pressure changes in the cylinder in real time. According to the change of atmospheric pressure, SADP can be very sensitive to liquid level detection, clot detection, bubble detection and other functions.

10.1 Liquid Level Detection

The liquid level detection function can be started by sending Ld commands through serial interface or CAN dictionary 0x4007 write operation. After the liquid level detection function is turned on, SADP will monitor the change of atmospheric pressure in real time, then Z-axis will start to drop, and when the pipette detects the change of atmospheric pressure, it will prompt the user to detect the liquid level through multiple ways.

The liquid level detection level output logic is as follows:

- ① When register 10 configuration location is 0, the pipette immediately outputs a high level voltage lasting 10 ms when it detects the liquid level.
- ② When register 10 configuration location is 1, the pipette outputs a low level voltage immediately when it receives the liquid level detection commands, and outputs a high level voltage immediately when it detects the liquid level.
- ③ When register 10 configuration location is 2, the pipette outputs a high level voltage immediately when it receives the liquid level detection commands and outputs a low level voltage immediately when it detects the liquid level.

Recommended process for liquid level detection:

When liquid level detection, drop the SADP that loaded with disposable tips to upper surface of the reagent vessel 5mm spacing firstly, then send the liquid level detection commands and continue to drop, and stop the Z-axis when the output signal of liquid level detection is received from non-trigger level to trigger level for 1 ms (the continuous 1 ms trigger level is to eliminate interference signals), or when the liquid level detection commands are received.

Caution:

- ① Detection of liquid level response time is about 30ms, so the recommended liquid level detection Z-axis drop velocity is less than 30mm/s
- ② There should be no foreign object (including residual reagents) at the tip of the disposable tips during liquid detection, otherwise the foreign object will slide down to the tip of the disposable tips when the Z-axis drop and the pipette cannot accurately detect the liquid level.
- ③ The original factory parameters only adapt to the liquid level detection of some disposable tips, if the user detects the liquid level and finds that the liquid level detection is triggered before it touches the liquid level, the liquid level detection coefficient will be increased appropriately, and if the detection is deep, the liquid level detection coefficient will be reduced appropriately

10.2 Reagent Abnormal Detection

In the process of aspiration, users will encounter in the aspirate clot, aspirate bubble, aspirate air and so on, we can detect these abnormalities through the change of atmospheric pressure, by default, these functions are closed, if you want to enable these functions can set register 60 to open the corresponding function, each bit corresponds to a function, the corresponding bit write 1 to enable 0 to close. Note that you need to convert hexadecimal to decimal before writing, for example, to enable the detection of aspirate bubbles and clots 60 register write 3 (hexadecimal 0x03).

Register address	Function
60	Pressure abnormality detection enable, HEX code, corresponding bit is 1 to enable the function Bit0: Aspiration of clot detection enable Bit2: Aspiration of air detection enable

Note: The aspiration abnormality detection will only be initiated by the pipette when the aspirate volume is not less than 20 μ L, because the aspirate volume is too small for the atmospheric pressure changes to be obvious. If there are reagents in the disposable tips to aspirate liquid, the aspiration of air and bubble detection are not reliable at this time. If the user needs to perform aspiration and dispense of air operation before normal aspiration and dispense of liquid, please set register 60 to turn off pressure abnormal detection first to prevent error report, and then turn on pressure abnormal detection when normal aspiration and dispense of liquid

11. Common Faults and Troubleshooting Methods

Note: The following descriptions are troubleshooting methods that arise when performed in a compliant operating environment and under rated operating conditions.

	Fault	Reason	Troubleshooting method
1	Input commands do not work	The drive has reported an error	Re-powering
2	Location is inaccurate or input does not match	Input error	Refer to the specification and re-enter the correct commands
		Signal interference	Check if the serial interface cables are in good condition
		Loose motor cable contact	Plugging the motor cable house tightly
		Rotation is prevented by external resistance	Troubleshoot the external resistance
		Not upgraded to the latest version of the program	Upgrade procedure (please consult your engineer if you need to upgrade)
3	Airflow is not working	Pipeline interface, pipeline blockage	Remove foreign objects or replace interfaces and pipelines

12. Environmental Conditions

Projects	Unit	Value
Working environment temperature	Degrees Celsius	+5°C~+40°C
Humidity of working environment	RH%	30%~80% non-condensing
Storage temperature	Degrees Celsius	-20°C~+60°C
Storage humidity	RH%	10%~85%

13. Safety Precautions

- For you and other users'safety, please read the safety precautions carefully.
- This manual uses the following marks. Please fully understand what they mean before reading on.



Any content with this mark, related to the safe use of the product and the user's personal safety, must be strictly in accordance with the requirements of the operation, otherwise it may cause damage to the product or endanger the user's personal safety.



Any content with this mark is a part of the user must pay attention to, otherwise it will cause damage to the product or other losses due to improper operation.



Please turn off the power when it is idle for a long time or when the whole machine is repaired, otherwise it will cause fire or electric shock.

Do not put it in wet, dusty, greasy environment or close to heat generating equipment, otherwise it will cause product failure, even malfunction, fire or electric shock.

Prohibit hot-swapping any serial interface cable, motor cable, optocoupler cable and valve power cable, otherwise it will cause communication or other parts to fail.

If there is a long-term non-use of the hole please use the matching plug plug into, otherwise may cause impurity and airflow into the valve body affect the normal use.

It is forbidden to disassemble the valve or adjust any parameters by yourself, otherwise the valve may not work properly.

Prohibit dry grinding operation without liquid infiltration.



Do not disassemble
Do not disassemble, repair or modify the product by yourself, otherwise it may cause fire or electric shock.

Avoid use in wet environments Moisture may cause electric shock.

Cut off the power when abnormal
If there is an abnormal situation, immediately cutoff the power. Otherwise, it may cause fire or electric shock.

Protection when using corrosive fluids
Strictly follow the applicability medium of the specification book to use, when using corrosive fluids must pay attention to protection.

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